

DIGITAL SAT MATH PRACTICE

Quadratic Word Problems

40 Questions | ~15% Grid Ins | Difficulty Labeled

TOPIC: QUADRATIC WORD PROBLEMS

Question 1

[Easy]

A ball is thrown upward from the ground. Its height h , in feet, after t seconds is modeled by $h(t) = -16t^2 + 48t$. What is the maximum height the ball reaches?

- A) 36 feet
- B) 48 feet
- C) 64 feet
- D) 96 feet

Question 2

[Easy]

A rectangular garden has a length that is 5 meters more than its width w . The area of the garden is 84 square meters. Which equation represents this situation?

- A) $w(w + 5) = 84$
- B) $w(w - 5) = 84$
- C) $2w + 2(w + 5) = 84$
- D) $w + (w + 5) = 84$

Question 3

[Easy]

A toy rocket's height, in meters, t seconds after launch is given by $h(t) = -5t^2 + 30t$. After how many seconds does the rocket return to the ground?

- A) 3
- B) 5
- C) 6
- D) 10

Question 4**[Easy]**

The product of two consecutive positive integers is 156. What is the smaller integer?

- A) 11
 - B) 12
 - C) 13
 - D) 14
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Question 5**[Easy]**

The function $f(t) = -16t^2 + 64t + 3$ models the height in feet of a ball t seconds after being kicked. What does the value 3 represent in this context?

- A) The maximum height of the ball
 - B) The time the ball hits the ground
 - C) The initial height of the ball when kicked
 - D) The speed of the ball at launch
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Question 6**[Medium]**

A farmer has 120 meters of fencing to enclose a rectangular pen against a barn wall (the barn forms one side). What is the maximum area the pen can have, in square meters?

- A) 900
 - B) 1200
 - C) 1500
 - D) 1800
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Question 7**[Medium]**

A company's daily profit P , in dollars, from selling x units of a product is modeled by $P(x) = -2x^2 + 100x - 300$. How many units should be sold to maximize profit?

- A) 20
- B) 25
- C) 30
- D) 50

Question 8**[Medium]**

A stone is launched upward from a 48-foot platform with an initial velocity of 32 feet per second. Its height is modeled by $h(t) = -16t^2 + 32t + 48$. After how many seconds does the stone hit the ground?

- A) 1
 - B) 2
 - C) 3
 - D) 4
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Question 9**[Medium]**

The revenue R , in thousands of dollars, from selling x hundred units of a gadget is $R(x) = -3x^2 + 36x - 5$. What is the maximum revenue?

- A) \$103,000
 - B) \$108,000
 - C) \$103,500
 - D) \$105,000
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Question 10**[Medium]**

A rectangle has a perimeter of 60 cm. Which of the following gives the area A of the rectangle as a function of its width w ?

- A) $A = w(30 - w)$
 - B) $A = w(60 - w)$
 - C) $A = w(60 - 2w)$
 - D) $A = w(15 - w)$
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Question 11**[Medium]**

A diver jumps from a 10-meter platform. Her height above the water is $h(t) = -4.9t^2 + 3.5t + 10$, where t is in seconds. How long does it take her to reach the water? (Round to the nearest tenth.)

Student Produced Response (Grid In)

Question 12**[Medium]**

A clothing store models its weekly profit with $P(x) = -5x^2 + 200x - 1500$, where x is the price in dollars. At what price does the store break even for the first time ($P = 0$)?

- A) \$10
 - B) \$15
 - C) \$20
 - D) \$30
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Question 13**[Medium]**

A ball is thrown from the top of a building. Its height h in feet after t seconds is $h(t) = -16t^2 + 32t + 80$. What is the height of the building?

- A) 32 feet
 - B) 48 feet
 - C) 80 feet
 - D) 96 feet
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Question 14**[Medium]**

The number of bacteria in a culture is modeled by $N(t) = 2t^2 + 12t + 50$, where t is the number of hours. How many bacteria are in the culture after 5 hours?

Student Produced Response (Grid In)

Question 15**[Medium]**

A swimming pool is being drained. The volume of water V , in gallons, remaining after t minutes is $V(t) = 2t^2 - 80t + 800$. After how many minutes is the pool completely empty?

- A) 10
- B) 20
- C) 30
- D) 40

Question 16**[Medium]**

The height of a water fountain stream is modeled by $h(x) = -0.5x^2 + 4x$, where x is the horizontal distance in meters. What is the horizontal distance from the fountain to where the water lands?

- A) 4
- B) 6
- C) 8
- D) 10

Question 17**[Hard]**

A theater charges \$12 per ticket and sells 200 tickets. For every \$1 increase in price, 5 fewer tickets are sold. What ticket price maximizes revenue?

- A) \$22
- B) \$26
- C) \$30
- D) \$32

Question 18**[Hard]**

Two positive numbers differ by 7, and their product is 198. What is the larger number?

Student Produced Response (Grid In)

Question 19**[Hard]**

A farmer uses 180 feet of fencing to build two identical rectangular pens side by side (sharing one wall). What is the maximum total area, in square feet?

- A) 1350
- B) 1800
- C) 2025
- D) 2700

Question 20**[Hard]**

A cannonball is fired at an angle. Its height h , in meters, at horizontal distance x meters is modeled by $h(x) = -0.01x^2 + 2x + 1$. What is the maximum height of the cannonball?

- A) 100 m
 - B) 101 m
 - C) 200 m
 - D) 201 m
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Question 21**[Hard]**

A store owner determines that the daily cost C (in dollars) to produce x items is $C(x) = 0.5x^2 - 20x + 350$. What is the minimum daily cost?

- A) \$100
 - B) \$150
 - C) \$200
 - D) \$250
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Question 22**[Hard]**

A landscaper designs a garden in the shape of a right triangle with legs of length x and $(x + 3)$ meters. If the area of the garden is 44 square meters, what is the length of the longer leg?

- A) 8
 - B) 11
 - C) 14
 - D) 16
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Question 23**[Hard]**

A model rocket's height in feet is $h(t) = -16t^2 + 96t + 5$. When does the rocket first reach a height of 149 feet?

- A) 2 seconds
- B) 3 seconds
- C) 4 seconds
- D) 5 seconds

Question 24**[Hard]**

An object's velocity v (in m/s) after t seconds is $v(t) = 3t^2 - 24t + 50$. At what time is the velocity at its minimum?

Student Produced Response (Grid In)

Question 25**[Hard]**

A cable hangs between two poles 40 meters apart. The height of the cable (in meters) at horizontal position x is modeled by $h(x) = 0.02x^2 - 0.8x + 15$, where x is the distance from the left pole. What is the minimum height of the cable?

- A) 5
 - B) 7
 - C) 8
 - D) 10
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Question 26**[Hard]**

A square's side length is increased by x cm, and a different side is decreased by x cm, forming a rectangle. The area of the rectangle is 91 cm^2 , and the original square had a side length of 10 cm. What is the value of x ?

- A) 1
 - B) 3
 - C) 5
 - D) 9
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Question 27**[Hard]**

A projectile's height is $h(t) = -16t^2 + 128t$. For how many seconds is the projectile above 240 feet?

- A) 1
- B) 2
- C) 3
- D) 4

Question 28**[Hard]**

A sports stadium's arch is modeled by $h(x) = -0.004x^2 + 1.6x$, where h is the height in feet and x is the horizontal distance in feet. What is the width of the arch at its base?

- A) 200 feet
- B) 300 feet
- C) 400 feet
- D) 500 feet

Question 29**[Hard]**

A phone company models monthly subscribers S (in thousands) with $S(t) = -0.5t^2 + 8t + 10$, where t is months since launch. During which month does the subscriber count first begin to decline?

- A) Month 6
- B) Month 8
- C) Month 9
- D) Month 10

Question 30**[Hard]**

In a physics experiment, the displacement d of a spring (in cm) at time t (in seconds) is $d(t) = 3t^2 - 18t + 24$. At what time(s) is the displacement equal to zero?

- A) $t = 2$ only
- B) $t = 4$ only
- C) $t = 2$ and $t = 4$
- D) $t = 3$ only

Question 31**[Hard]**

A bridge's support cable follows the curve $y = 0.005x^2 - 3x + 500$, where y is the height in meters and x is the horizontal distance. What is the lowest point of the cable?

- A) 25 m
- B) 50 m
- C) 75 m
- D) 100 m

Question 32**[Hard]**

A company's profit function is $P(x) = -2x^2 + 120x - 1000$, where x is the number of units sold. For what range of units sold is the company profitable ($P > 0$)?

- A) Between 10 and 50 units
 - B) Between 5 and 55 units
 - C) Between 20 and 40 units
 - D) Between 15 and 45 units
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Question 33**[Hard]**

An athlete throws a javelin. Its height h (in meters) at horizontal distance x (in meters) is $h(x) = -0.02x^2 + 1.4x + 2$. What is the horizontal distance when the javelin hits the ground? (Round to the nearest whole number.)

Student Produced Response (Grid In)

Question 34**[Hard]**

A rectangular lot has an area of 180 m^2 . If the length is decreased by 3 m and the width is increased by 2 m, the area remains 180 m^2 . What are the original dimensions?

- A) 12 m by 15 m
 - B) 10 m by 18 m
 - C) 9 m by 20 m
 - D) 6 m by 30 m
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Question 35**[Hard]**

The height of a firework t seconds after launch is $h(t) = -16t^2 + 128t$. What is the time interval during which the firework is higher than 240 feet?

- A) From $t = 3$ to $t = 5$
- B) From $t = 2$ to $t = 6$
- C) From $t = 3$ to $t = 6$
- D) From $t = 2$ to $t = 5$

Question 36**[Hard]**

A manufacturer finds that the number of defective items per batch follows $D(x) = 0.1x^2 - 4x + 50$, where x is the production speed. At what speed is the number of defects minimized, and what is that minimum?

- A) Speed 20, minimum 10 defects
- B) Speed 20, minimum 15 defects
- C) Speed 40, minimum 10 defects
- D) Speed 10, minimum 20 defects

Question 37**[Hard]**

A tennis ball is hit at a height of 2 feet. Its height $h(t) = -16t^2 + 40t + 2$. A camera is positioned to photograph the ball when it is exactly at its maximum height. At what time should the camera take the photo, and what height will the ball be at?

Student Produced Response (Grid In)

Question 38**[Hard]**

The path of a golf ball is modeled by $h(x) = -x^2/200 + x$, where h is the height in yards and x is the horizontal distance. The ball must clear a 10-yard-tall tree located 40 yards from the tee. Does the ball clear the tree, and by how much?

- A) Yes, by 22 yards
- B) Yes, by 12 yards
- C) No, it falls short by 2 yards
- D) Yes, by 32 yards

Question 39**[Hard]**

A city's population P (in thousands) t years after 2020 is modeled by $P(t) = -0.3t^2 + 6t + 50$. During what year does the population reach its maximum?

- A) 2028
- B) 2030
- C) 2032
- D) 2035

Question 40**[Hard]**

A rectangular frame is made from 56 cm of wire. The frame must enclose the largest possible area. What is the maximum area, in cm^2 ?

Student Produced Response (Grid In)